

Optimizer comparison for fully connected neural networks

Programming Assignment 3

Group 32

Autumn semester 2026

Reproduction of the supplied Assignment 3 specification on the original Group 32 MNIST splits. This newly measured study supplies the Assignment 3 baseline required by Assignment 4.

Component	Coverage
Architectures	Three, four, and five ReLU hidden layers; varied intermediate widths
Optimizers	SGD, batch GD, momentum, NAG, AdaGrad, RMSProp, Adam
Stopping	Absolute successive epoch mean training loss difference below 10^{-4}
Model selection	Highest validation accuracy across converged architecture/optimizer runs

All runs use the prescribed learning rate, batch sizes, and optimizer parameters. There is no epoch limit. All measured tables and figures are saved with the source code.

1 Dataset and reproducible protocol

Digit	Train	Validation	Test
0	2277	759	759
3	2277	759	759
5	2277	759	759
6	2277	759	759
9	2277	759	759
Total	11385	3795	3795

The supplied 28×28 grayscale images are flattened to 784 pixels and scaled to $[0, 1]$ by dividing by 255. The supplied splits are retained without re-splitting. Labels are digits 0, 3, 5, 6, and 9. No additional per-pixel standardization is applied in this Assignment 3 study.

Name	Hidden layers	Full architecture
3_layers	3	$784 \rightarrow 32 \rightarrow 16 \rightarrow 8 \rightarrow 5$
4_layers	4	$784 \rightarrow 32 \rightarrow 24 \rightarrow 16 \rightarrow 8 \rightarrow 5$
5_layers	5	$784 \rightarrow 32 \rightarrow 24 \rightarrow 16 \rightarrow 12 \rightarrow 8 \rightarrow 5$

The first hidden width is held at 32; depth and intermediate widths vary. ReLU is used in each hidden layer, with five linear logits and softmax cross-entropy. Initialization is seeded He normal with zero biases. For each architecture, every optimizer starts from identical weights and biases, verified using a SHA-256 digest. Online optimizers also use the same seeded image permutation in corresponding epochs.

$$L(\theta) = \frac{1}{N} \sum_{i=1}^N -\log \frac{\exp z_{i,y_i}}{\sum_{c=1}^5 \exp z_{i,c}}.$$

The stopping rule is applied to the complete training split at the end of each epoch, after that epoch's updates: $|L_t - L_{t-1}| < 10^{-4}$. Training stops at the first qualifying epoch; the first possible stop is epoch 2. Validation does not stop training or choose an epoch. Final converged weights are evaluated on validation for model selection.

Numba compiles the exact sample-wise forward, backward, and parameter update loops in float64. Each online sample is followed by one update; it is not replaced by minibatch training. The full-batch implementations use NumPy matrix products. All seven update rules passed numerical comparisons with PyTorch.

2 Prescribed optimizer settings

Optimizer	Batch size	Parameters
SGD	1	$\eta = 0.001$
Batch gradient descent	11385	$\eta = 0.001$; mean full-batch gradient
SGD with momentum	1	$\eta = 0.001, \mu = 0.9$
Nesterov accelerated gradient	1	$\eta = 0.001, \mu = 0.9$
AdaGrad	11385	$\eta = 0.001, \epsilon = 10^{-8}$; zero initial accumulator
RMSProp	11385	$\eta = 0.001, \rho = 0.99, \epsilon = 10^{-8}$
Adam	1	$\eta = 0.001, \beta_1 = 0.9, \beta_2 = 0.999, \epsilon = 10^{-8}$

Momentum uses $v_t = 0.9v_{t-1} + g_t$ and $\theta_t = \theta_{t-1} - 0.001v_t$. NAG uses the equivalent update convention in PyTorch SGD with `nesterov=True`: $\theta_t = \theta_{t-1} - 0.001(g_t + 0.9v_t)$. RMSProp is uncen-tered with no additional momentum, and Adam uses bias correction. No weight decay, dropout, or learning-rate schedule is introduced.

3 Epochs to the specified stopping threshold

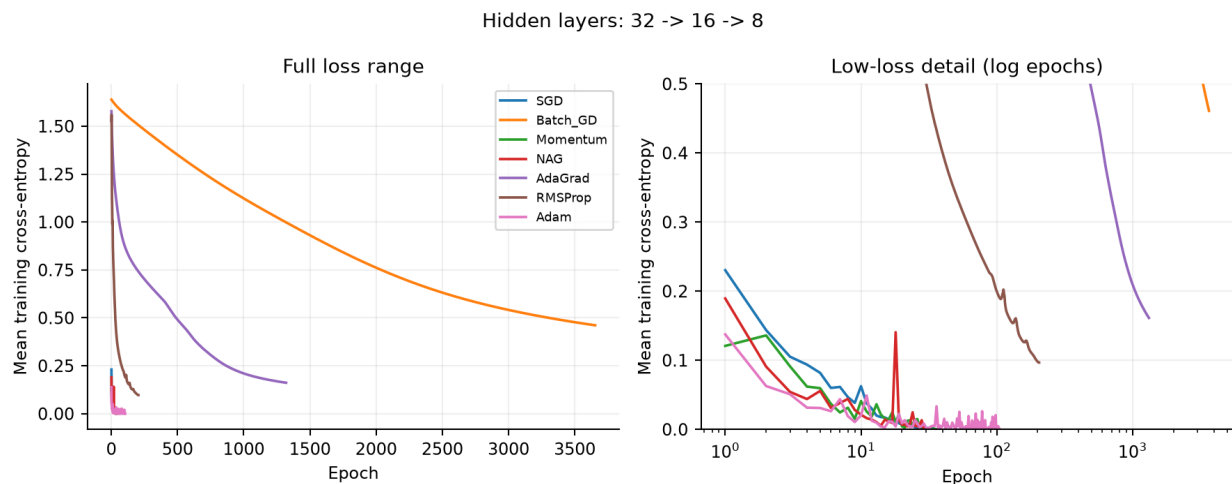
Architecture	SGD	Batch_GD	Momentum	NAG	AdaGrad	RMSProp	Adam
3_layers	34	3654	37	32	1320	205	105
4_layers	17	3957	36	30	1397	146	118
5_layers	20	3448	42	56	1091	202	15

The table counts complete passes through the training set. One online epoch contains 11,385 optimizer updates, whereas one full-batch epoch contains one update. Consequently, epochs alone are not a matched computational budget. Every run's final absolute loss difference is below 10^{-4} , and no earlier recorded pair satisfies the stopping condition.

This literal stopping rule can classify a tiny loss change as convergence even at a high loss. In particular, a full-batch learning rate of 0.001 can produce small changes near initialization. The report preserves such outcomes instead of continuing beyond the assigned criterion or substituting a fixed epoch budget.

4 Results: 32 - 16 - 8

Optimizer	Epochs	Train acc.	Val. acc.	Final CE	Final loss delta
SGD	34	100.00%	97.29%	0.001321	9.41e-05
Batch_GD	3654	87.58%	87.91%	0.461011	1.00e-04
Momentum	37	99.98%	97.89%	0.000334	5.49e-05
NAG	32	99.99%	97.89%	0.000289	5.21e-05
AdaGrad	1320	95.69%	94.68%	0.161278	9.99e-05
RMSProp	205	97.16%	96.05%	0.096727	4.02e-05
Adam	105	100.00%	97.87%	0.000000	2.04e-06

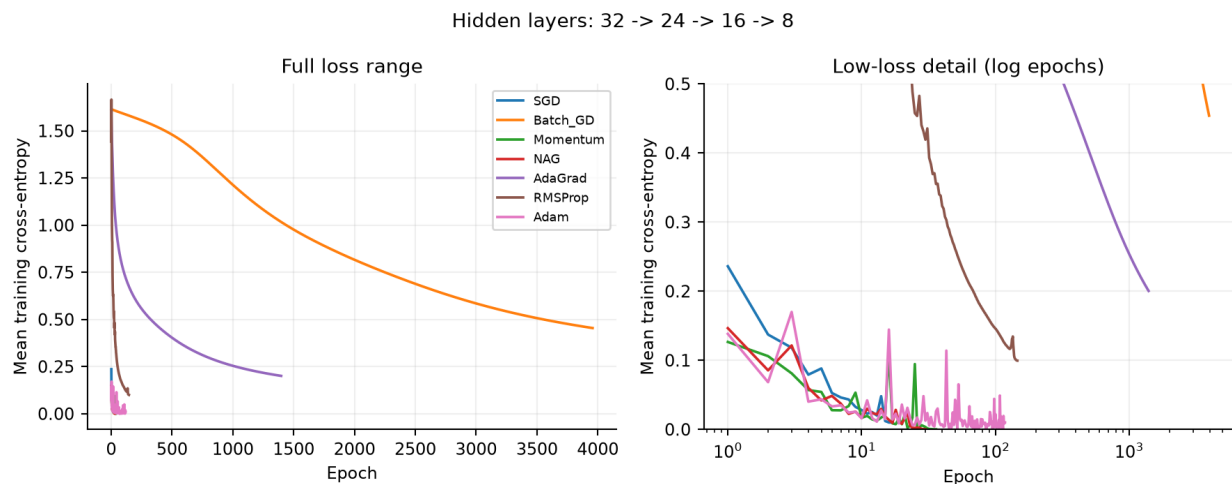


Superimposed post-epoch average training cross-entropy for all seven optimizers. The right panel magnifies losses below 0.5 with a logarithmic epoch axis; high-loss curves remain visible in the left panel.

For this architecture, NAG has the highest validation accuracy, 97.89%. The full curve, convergence loss difference, number of updates, and measured runtime are available in `metrics.json`.

5 Results: 32 - 24 - 16 - 8

Optimizer	Epochs	Train acc.	Val. acc.	Final CE	Final loss delta
SGD	17	99.81%	96.89%	0.009842	9.38e-05
Batch_GD	3957	86.42%	85.93%	0.453977	1.00e-04
Momentum	36	100.00%	97.81%	0.000085	2.29e-05
NAG	30	99.99%	97.89%	0.000288	7.03e-05
AdaGrad	1397	94.86%	93.62%	0.200351	9.97e-05
RMSProp	146	97.12%	95.73%	0.099436	4.15e-05
Adam	118	99.82%	97.52%	0.009932	6.13e-05

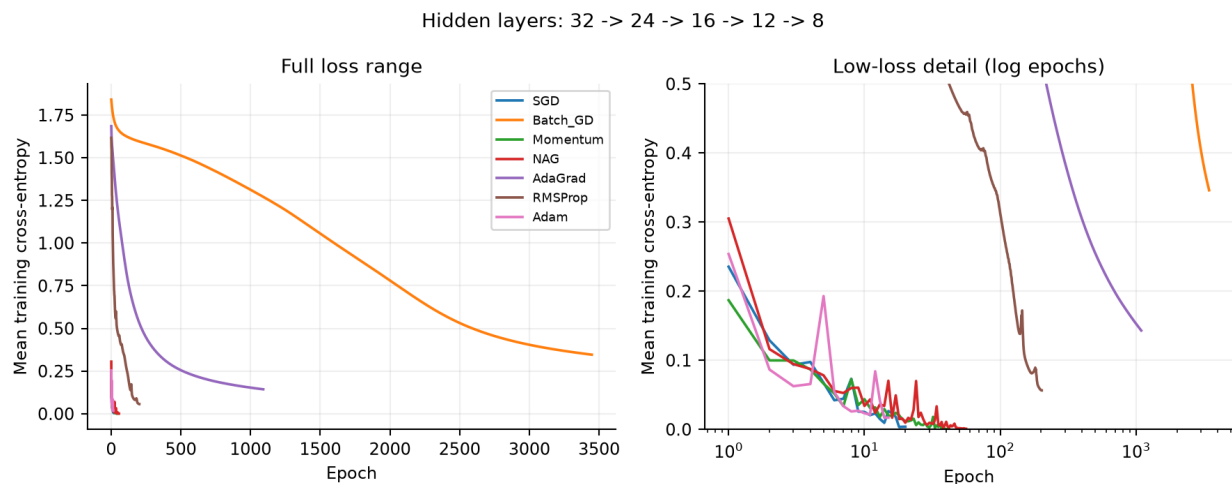


Superimposed post-epoch average training cross-entropy for all seven optimizers. The right panel magnifies losses below 0.5 with a logarithmic epoch axis; high-loss curves remain visible in the left panel.

For this architecture, NAG has the highest validation accuracy, 97.89%. The full curve, convergence loss difference, number of updates, and measured runtime are available in `metrics.json`.

6 Results: 32 - 24 - 16 - 12 - 8

Optimizer	Epochs	Train acc.	Val. acc.	Final CE	Final loss delta
SGD	20	99.93%	97.31%	0.003807	9.63e-05
Batch_GD	3448	89.75%	89.04%	0.346000	9.99e-05
Momentum	42	99.99%	97.55%	0.000303	3.27e-05
NAG	56	99.96%	97.52%	0.000860	6.70e-05
AdaGrad	1091	96.19%	94.73%	0.143059	9.99e-05
RMSProp	202	98.65%	96.31%	0.056387	3.71e-06
Adam	15	99.52%	97.52%	0.016955	5.36e-05

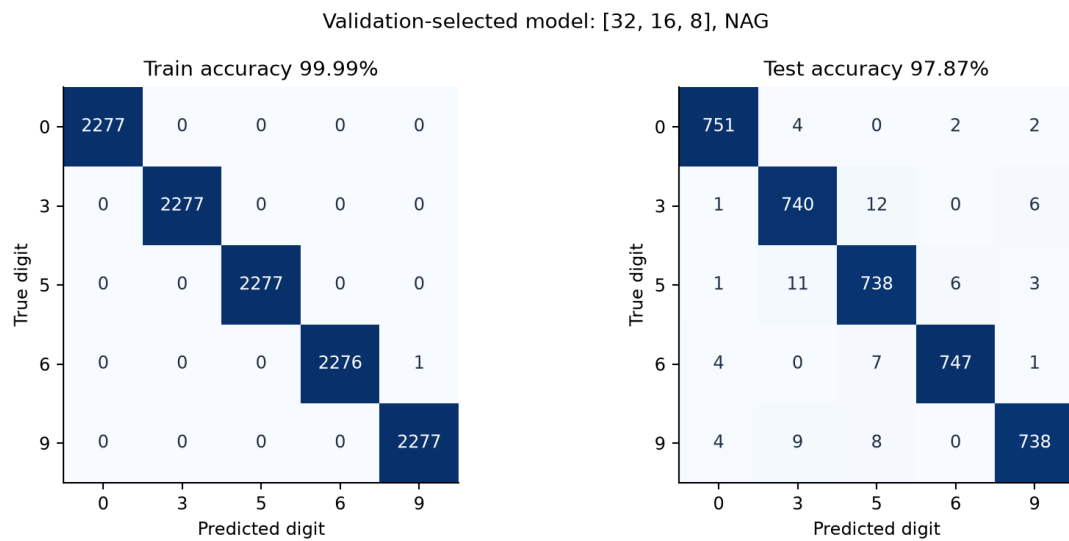


Superimposed post-epoch average training cross-entropy for all seven optimizers. The right panel magnifies losses below 0.5 with a logarithmic epoch axis; high-loss curves remain visible in the left panel.

For this architecture, Momentum has the highest validation accuracy, 97.55%. The full curve, convergence loss difference, number of updates, and measured runtime are available in `metrics.json`.

7 Validation-selected model and test evaluation

Attribute	Selected result
Hidden layers	32 $\rightarrow$ 16 $\rightarrow$ 8
Optimizer	NAG
Convergence epochs	32
Training accuracy	99.99%
Validation accuracy	97.89%
Test accuracy	97.87%



Rows are true digits and columns are predicted digits. The training matrix has 2,277 observations per row; the test matrix has 759 per row.

The largest directional test confusion is digit 3 predicted as 5 (12 images). The test split is used only after the final architecture and optimizer have been selected on validation accuracy; validation cross-entropy breaks an accuracy tie.

8 Observations and reproducibility

The comparison combines different update schedules because the brief explicitly prescribes them. Any performance gap therefore reflects both the optimizer and its assigned batch size. Identical initialization controls the starting point, but does not remove those scheduling differences. Small epoch-to-epoch loss changes also do not prove a minimum or high classification accuracy.

There is one seeded run per architecture and optimizer, not repeated trials. The tested compact widths are a finite architecture search, so the selected result is the best among these experiments. It is a fresh reproduction of Assignment 3, not a claim about a previously submitted model.

Source: `CS601T_Assignment3_Details_07Sept2026.pdf`. The handout's printed date contains a year inconsistency; it does not change the experiment requirements.

Code folder: `Group32.Assignment3_code`.

Run `python run_assignment3.py`, then `python verify_results.py`. The measured comparison record is `results/assignment3_best.json`.

Dataset SHA-256: `d973de54462a8f8710f74cb65d7b5670993802b9252e531d78aaee19d93cb2a8`.